

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

MAY/JUNE EXAMINATION

BMIT3084 ENTERPRISE NETWORKING

TUESDAY, 27 MAY 2025

TIME: 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN INFORMATION
SECURITY

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN INTERNET
TECHNOLOGY

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BMIT3084 ENTERPRISE NETWORKING**Question 1**

You have been employed as a network administrator to analyse, illustrate and configure various types of static routes for a network topology with Internet Protocol version 4 (IPv4) addressing configured on all devices as shown in Figure 1-1.

Assume a **fully-specified default static route** is pre-configured in **SUMMER** and also a **default static route** is pre-configured in **SPRING** to forward traffic to the **ISP** and the Internet. In addition, **standard static routes** were pre-configured in the **ISP** to forward packets to **WINTER_LAB** and **SUMMER_LAB**. Refer to Figure 1-1 and answer the following questions.

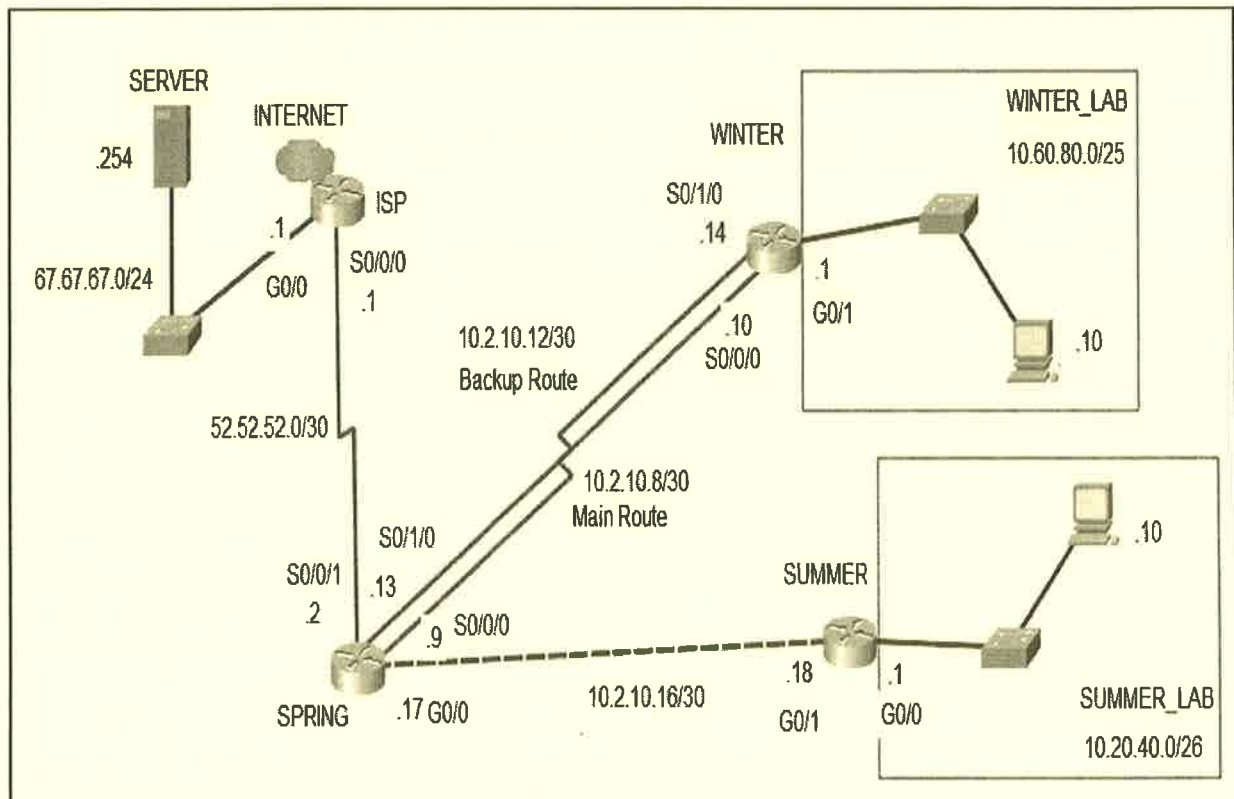


Figure 1-1: A network topology

- a) (i) Configure a **default static route** and a **floating default static route** in **WINTER** using **next-hop IP address** to forward traffic to **SPRING**. State your assumptions for the **floating default static route**. (4 marks)
- (ii) Configure a **standard static route** and a **floating standard static route** using **next-hop IP address** in **SPRING** to forward packets to **WINTER_LAB**. State your assumptions for the **floating standard static route**. (4 marks)
- (iii) Configure a **fully-specified standard static route** in **SPRING** to forward packets to **SUMMER_LAB**. (4 marks)
- (iv) Why is a **fully-specified standard static route** implemented in Question 1 a) (iii)? Justify your answers. (4 marks)

Question 1 (Continued)

- b) A network topology in Figure 1-2 was pre-configured with a **default static route** in **CENTRE** to forward traffic to the **ISP** and the **Internet**. In addition, the **ISP** is already pre-configured with static routing. You are required to implement **Open Shortest Path First (OSPF)** configurations using the network command with wildcard mask based on the subnet mask in **CENTRE** and **SOUTH** only. Propagate the default route in **CENTRE** to **NORTH** and **SOUTH** to forward traffic to the **ISP** and the **Internet**. Use OSPF **process-id 73** and **area-id 0**. Use Table 1-1 to document your answer.

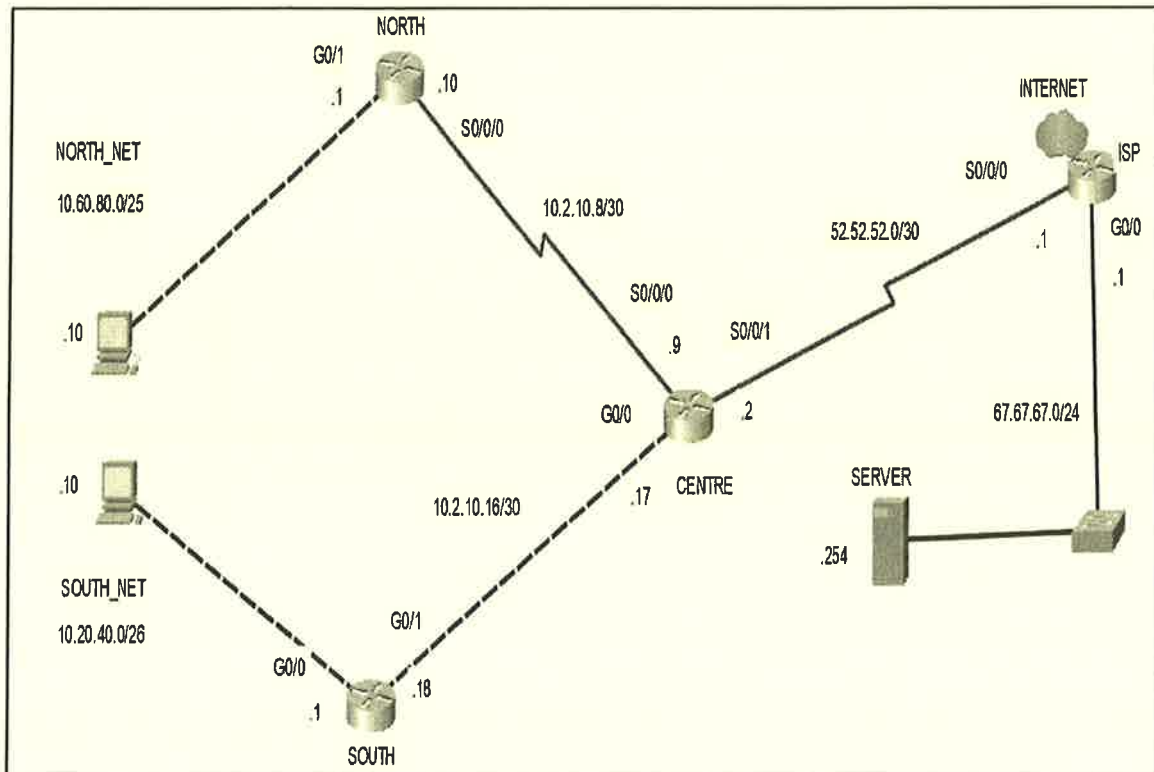


Figure 1-2: A network topology

Table 1-1: Documentation Table

Router name	Configurations

(9 marks)

[Total: 25 marks]

Question 2

- a) (i) Internet Control Message Protocol (ICMP) is a network-layer Internet protocol. Explain **THREE (3)** ICMP messages used by threat actors. (6 marks)
- (ii) Propose **TWO (2)** methods to prevent ICMP attacks by threat actors. (5 marks)

BMIT3084 ENTERPRISE NETWORKING**Question 2 (Continued)**

- b) Access Control Lists (ACLs) are used to filter network traffic. Based on Figure 2-1, answer the following questions. Assume all hosts and routers can communicate with each other.

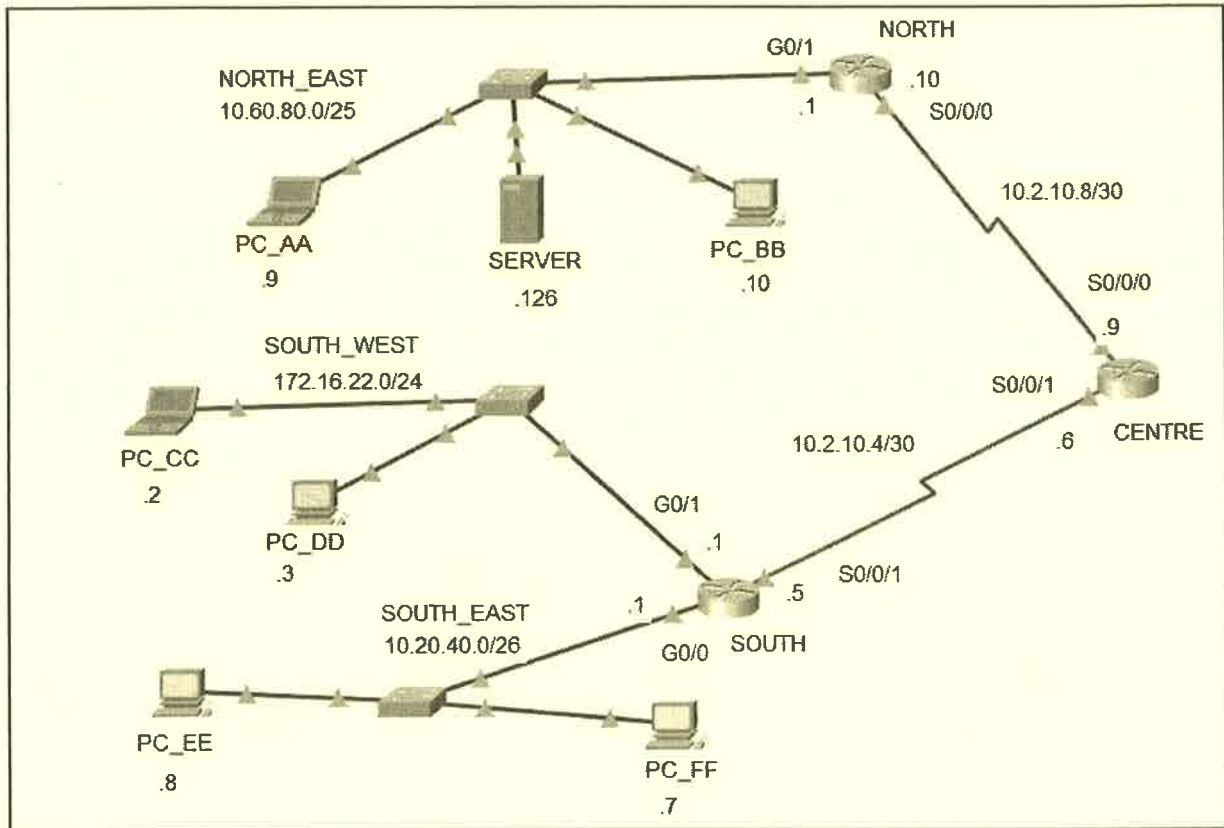


Figure 2-1: A network topology

Examine Figure 2-1 and write an extended access list named **CONTROL_TRAFFIC** to allow odd numbered IP addresses from the **SOUTH_WEST** network to access the **SERVER** via **HTTP**. Allow the first seven usable IP addresses from the **SOUTH_EAST** network to ping the **NORTH_EAST** network. Deny all other traffic, which must be explicitly written in your ACL. Use keywords in your ACL, except for port number **80** for **HTTP**. Indicate the **router**, **interface** and **direction** to apply the ACL efficiently.

(14 marks)

[Total: 25 marks]

BMIT3084 ENTERPRISE NETWORKING**Question 3**

Dynamic Host Configuration Protocol (DHCP), Static Network Address Translation (NAT) and Port Address Translation (PAT) configurations are to be configured in Figure 3-1 network topology. Assume OSPF and static routing protocols are configured in the respective routers.

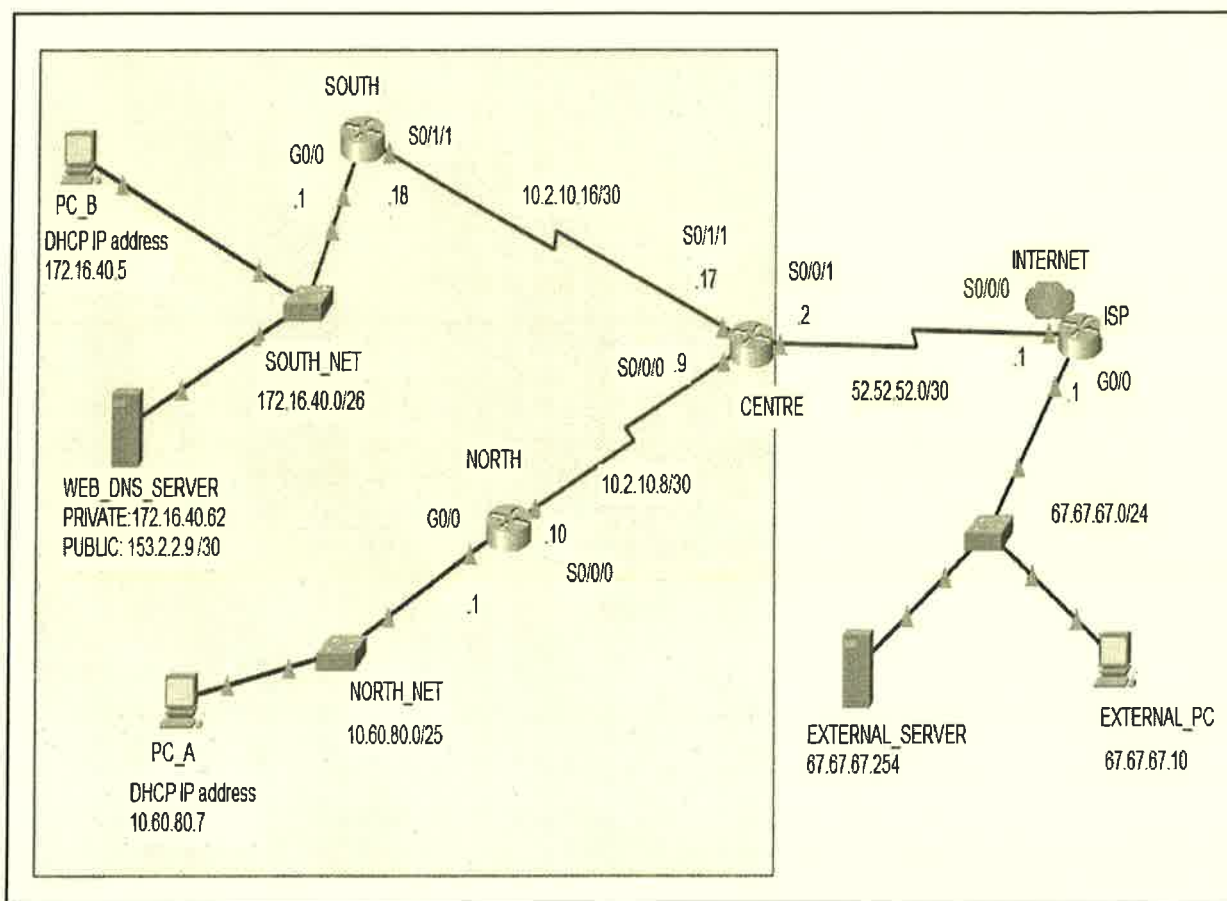


Figure 3-1: A network topology

SOUTH	NORTH
ip dhcp excluded-address 172.16.40.1 172.16.40.4 ip dhcp excluded-address 10.60.80.1 10.60.80.6	interface GigabitEthernet0/0 ip address 10.60.80.1 255.255.255.128
ip dhcp pool DHCP_SOUTH_NET network 10.20.40.5 255.255.255.192 dns-server 172.16.40.62	interface Serial0/0/0 ip address 10.2.10.10 255.255.255.252
ip dhcp pool DHCP_NORTH_NET dns-server 172.16.40.62	
interface GigabitEthernet0/0 ip address 172.16.40.1 255.255.255.192	
interface Serial0/1/1 ip address 10.2.10.18 255.255.255.252	

BMIT3084 ENTERPRISE NETWORKING**Question 3 (Continued)**

CENTRE
interface Serial0/0/0 ip address 10.2.10.9 255.255.255.252
interface Serial0/0/1 ip address 52.52.52.2 255.255.255.252
interface Serial0/1/1 ip address 10.2.10.17 255.255.255.252
ip nat pool CENTRE_NAT 153.2.2.9 153.2.2.9 netmask 255.255.255.252 access-list 99 permit 172.16.40.0 0.0.0.63

Figure 3-2: Partial output of “show run” command

- a) Analyse the partial output of “show run” command in Figure 3-2 and also the network topology in Figure 3-1. The **SOUTH** router is configured as a DHCP server. PC_A and PC_B are having problems obtaining the IP addresses and other DHCP configurations successfully. Use Table 3-1 to document all errors, provide the solutions or correct configurations for the respective errors and justify your answers. State your assumptions in your answers. (10 marks)

Table 3-1: Documentation Table

Errors	Solutions	Justifications

- b) (i) The public address **153.2.2.9** is assigned to the **WEB_DNS_SERVER** from the external network address **153.2.2.8/30**. Implement **static NAT** configurations on the appropriate router and apply them to all the router’s interfaces for the **WEB_DNS_SERVER** to be reachable from the Internet. (5 marks)
- (ii) Assume the static NAT configurations in Question 3 b) (i) have been successfully implemented. You are required to identify errors and provide solutions for **PAT** configurations using the **remaining** public IP address from the external network address **153.2.2.8/30** as the address pool. Use standard numbered access-list **99**. All internal PCs should be able to ping the **EXTERNAL_PC**. Use Table 3-2 to document your answers.

Table 3-2: Documentation Table

Errors	Solutions

(10 marks)
[Total: 25 marks]

BMIT3084 ENTERPRISE NETWORKING**Question 4**

- a) (i) A Wide Area Network (WAN) is owned by a service provider. A user must pay a fee to use the provider's network services to connect to remote sites. A single-carrier connection and a dual-carrier connection are two types of WAN carrier connections. Differentiate these **TWO (2)** WAN carrier connections with the aid of diagrams. (6 marks)
- (ii) Suggest **ONE (1)** disadvantage for each of the single-carrier and dual-carrier connections. (4 marks)
- b) Enterprise managed Virtual Private Networks (VPNs) can be deployed in two configurations. Distinguish the **TWO (2)** types of enterprise managed VPNs. (6 marks)
- c) List **THREE (3)** tools for implementing Quality of Service (QoS) and give **ONE (1)** explanation for each of the tools. (9 marks)
- [Total: 25 marks]